

Ticket 02 Study Report: Finite-Difference Grid and Black-Scholes Operator Setup

SURF2026 American Option Risk Surfaces

Codex-assisted implementation report

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Abstract

Ticket 2 builds the first numerical layer beneath the future option solver: uniform spot and time-to-maturity grids, an explicit interior-node convention, central finite-difference coefficients for the Black-Scholes spatial operator, a small operator-application helper, and simple boundary-value helpers. This report is written for beginner researchers with accounting, data, machine-learning, and Python experience who are still learning finite differences, PDE-based pricing, CN/PSOR, and American option free-boundary logic. Ticket 2 prepares later European Crank-Nicolson validation and American CN/PSOR work, but it does not implement or validate those solvers.

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1 Role of Ticket 2 in the Whole Solver Project

Ticket 1 gave the project analytic option-pricing utilities: payoff functions and European Black-Scholes closed-form prices. Ticket 2 begins the numerical side. Its purpose is to define the finite grid on which a future solver will approximate option values, and to define the discrete spatial operator that will later appear inside the Crank-Nicolson matrices.

This order matters because CN/PSOR combines several ideas at once: a grid, finite differences, time stepping, linear algebra, an American payoff obstacle, and iterative projection. If all of those pieces are written together, debugging becomes confusing. Ticket 2 isolates only the grid and the spatial operator. Later tickets can then test time stepping and projection against a known, reviewed numerical foundation.

In plain language, Ticket 2 answers three basic questions:

- Where are the spot and time points?
- Which spot nodes are solved by the PDE operator, and which nodes are imposed boundaries?
- What are the lower, diagonal, and upper finite-difference weights for the Black-Scholes spatial operator?

It is important to state the boundary of this ticket. Ticket 2 does not march through time, does not solve a linear system, does not run PSOR, does not enforce the American obstacle, and does not extract a free boundary. It creates reusable pieces that Ticket 3 can use for the European Crank-Nicolson validation check.

2 Theory and Methodology Connection

The source basis for this ticket is internal and project-specific:

- `reports/01_literature/literature_map.md`;
- `reports/02_math/formulation_note.md`;
- `reports/03_solver/solver_validation_plan.md`;
- `reports/03_solver/solver_implementation_tickets.md`;
- the Ticket 1 Black-Scholes utility report and code.

The literature map places this work in the classical finite-difference American option literature. It mentions Brennan-Schwartz finite-difference valuation as background and emphasizes that a trusted classical benchmark must come before neural surrogate modelling. This Ticket 2 report does not add a standalone bibliography because the available project notes do not provide complete bibliographic details for every external source. Instead, it uses the internal literature map and formulation note as the documented source basis.

The formulation note defines the dividend-adjusted Black-Scholes spatial operator:

$$LU = \frac{1}{2}\sigma^2 S^2 U_{SS} + (r - q)SU_S - rU. \quad (1)$$

The solver validation plan states that the finite-difference version L_h should approximate U_S and U_{SS} on interior nodes, while boundary conditions must be handled consistently later. Ticket 2 implements this L_h coefficient setup only. The future CN step will use

$$A = I - \frac{\Delta\tau}{2}L_h, \quad b^n = \left(I + \frac{\Delta\tau}{2}L_h\right)U^n, \quad (2)$$

but Ticket 2 deliberately does not build A , b^n , or any time-marching solver.

3 Finite-Difference Grids in Plain Language

A continuous option-pricing problem has infinitely many possible spot prices and times. A computer cannot solve infinitely many points directly. A finite-difference method replaces the continuous domain with a finite list of grid points.

For spot, Ticket 2 uses a uniform grid on a truncated domain:

$$S_i = i\Delta S, \quad i = 0, 1, \dots, M, \quad \Delta S = \frac{S_{\max}}{M}. \quad (3)$$

The domain starts at $S = 0$ and ends at $S = S_{\max}$. The true Black-Scholes model allows unbounded spot prices, but a numerical solver must choose a finite upper boundary. Later validation work will check whether $S_{\max}$ is large enough.

For time-to-maturity, Ticket 2 uses

$$\tau_n = n\Delta\tau, \quad n = 0, 1, \dots, N, \quad \Delta\tau = \frac{T}{N}. \quad (4)$$

The project uses τ , time-to-maturity, rather than calendar time. At $\tau = 0$, the option is at maturity and the value is the payoff. As τ increases, the future solver moves away from maturity toward earlier calendar times.

Uniform grids are not the most advanced choice, but they are the right first step here. They make indexing readable, make central finite-difference formulas simple, and keep the beginner implementation auditable. Nonuniform grids can be considered later only after the uniform-grid solver is validated.

4 Black-Scholes Spatial Operator

The operator in Equation 1 has three parts:

Term	Plain-language meaning
$\frac{1}{2}\sigma^2 S^2 U_{SS}$	Diffusion or curvature term. It captures how volatility and option curvature affect value.
$(r - q)SU_S$	Drift term. It captures the risk-neutral movement of the underlying after continuous dividends.
$-rU$	Discounting term. It subtracts the risk-free discounting effect from the value.

Ticket 2 converts this continuous operator into a tridiagonal discrete operator. “Tridiagonal” means that at an interior node i , the approximation uses only three neighboring values: U_{i-1} , U_i , and U_{i+1} .

5 Central Finite Differences

At an interior node S_i , central differences approximate the first and second spot derivatives:

$$U_S(S_i) \approx \frac{U_{i+1} - U_{i-1}}{2\Delta S}, \quad (5)$$

$$U_{SS}(S_i) \approx \frac{U_{i+1} - 2U_i + U_{i-1}}{(\Delta S)^2}. \quad (6)$$

Substituting these formulas into Equation 1 gives

$$(L_h U)_i = a_i U_{i-1} + b_i U_i + c_i U_{i+1}, \quad (7)$$

where

$$a_i = \frac{1}{2}\sigma^2 \frac{S_i^2}{(\Delta S)^2} - \frac{1}{2}(r - q) \frac{S_i}{\Delta S}, \quad (8)$$

$$b_i = -\sigma^2 \frac{S_i^2}{(\Delta S)^2} - r, \quad (9)$$

$$c_i = \frac{1}{2}\sigma^2 \frac{S_i^2}{(\Delta S)^2} + \frac{1}{2}(r - q) \frac{S_i}{\Delta S}. \quad (10)$$

The implementation calls these arrays `lower`, `diagonal`, and `upper`. The names come from the matrix form: a_i multiplies the value to the left, b_i multiplies the current node, and c_i multiplies the value to the right.

Two manufactured-function checks are useful:

$$L(1) = -r, \quad (11)$$

$$L(S) = -qS. \quad (12)$$

These identities are simple but powerful. They check the signs of the drift and discount terms without needing a full option solver.

6 Interior Nodes and Boundary Nodes

For a grid with nodes $0, \dots, M$, Ticket 2 defines:

- node 0 as the lower boundary at $S = 0$;
- node M as the upper boundary at $S = S_{\max}$;
- nodes $1, \dots, M - 1$ as interior nodes.

The function `interior_indices(M)` returns exactly those interior indices. This is a small helper, but it prevents a common off-by-one error. The future PDE solve should operate on interior nodes while boundary values are imposed separately.

Boundary values are included in the full value vector because the discrete operator at the first and last interior nodes needs U_0 and U_M . However, the operator helper returns only interior results. This keeps the distinction between imposed boundary values and solved interior values explicit.

7 Design Decisions

7.1 Uniform Grid First

The implementation uses uniform spot and time grids because the formulation note and solver validation plan introduce the method with ΔS and $\Delta \tau$. Uniform grids make the coefficient formulas easy to inspect and test. This is especially important for greenhand researchers who are still learning how the PDE becomes code.

7.2 Explicit Interior Indexing

Interior indexing is implemented as a named helper instead of being hidden inside slices throughout the solver code. The goal is not abstraction for its own sake. The goal is to make it obvious that nodes 0 and M are boundaries while nodes $1, \dots, M - 1$ are the PDE nodes.

7.3 Coefficient Arrays Instead of Full Matrix

Ticket 2 returns lower, diagonal, and upper coefficient arrays rather than a dense matrix. This is enough for tridiagonal finite-difference work and avoids introducing matrix assembly details before Ticket 3. The future CN solver can decide how to build or solve tridiagonal systems.

7.4 Simple Boundary Helpers

The boundary helpers return boundary values only. They do not adjust a Crank-Nicolson right-hand side, solve a PDE, or extract a free boundary. The helpers are included because later European validation and American cases need consistent boundary formulas.

8 Implementation Record

8.1 Files Created

File	Role in Ticket 2
<code>src/american_risk_surfaces/solvers/grid.py</code>	Ticket 02 module containing uniform spot/time grids plus explicit interior-node indices.
<code>src/american_risk_surfaces/solvers/operator.py</code>	Ticket 02 module containing Black-Scholes operator coefficients, operator application, and boundary helpers.
<code>tests/test_grid_operator.py</code>	Focused <code>unittest</code> coverage for grids, coefficients, manufactured-function checks, and boundaries.
<code>reports/03_solver/tickets/ticket_02_grid_operator.tex</code>	Formal LaTeX study-report source.
<code>reports/03_solver/tickets/ticket_02_grid_operator.pdf</code>	Formal PDF compiled from the LaTeX source.

9 Code-Level Function Explanations

Function or class	Purpose and important behavior
<code>uniform_spot_grid</code>	Builds $S_i = i\Delta S$ for $i = 0, \dots, M$. It validates $S_{\max} > 0$ and $M \geq 2$.
<code>uniform_tau_grid</code>	Builds $\tau_n = n\Delta\tau$ for $n = 0, \dots, N$. It validates $T \geq 0$ and $N \geq 1$. $T = 0$ is allowed.
<code>interior_indices</code>	Returns $1, \dots, M - 1$, making the boundary/interior convention explicit.
<code>BlackScholesOperatorCoefficients</code>	Frozen dataclass holding lower, diagonal, upper, and interior spot arrays.
<code>black_scholes_operator_coefficients</code>	Computes a_i , b_i , and c_i on interior nodes using central differences. It rejects non-increasing, nonuniform, or dS -mismatched spot grids.
<code>apply_black_scholes_operator</code>	Applies the tridiagonal operator to a full value vector and returns interior values only.
<code>european_put_boundaries</code>	Returns $(Ke^{-r\tau}, 0)$, the simple European put boundary pair.
<code>american_put_boundaries</code>	Returns $(K, 0)$, the simple American put boundary pair.
<code>european_call_boundaries</code>	Returns $(0, \max(S_{\max}e^{-q\tau} - Ke^{-r\tau}, 0))$.
<code>american_call_boundaries</code>	Returns $(0, \max(0, S_{\max} - K, S_{\max}e^{-q\tau} - Ke^{-r\tau}))$.

All helpers validate the domain inputs needed for this ticket: positive strike, nonnegative τ , positive $S_{\max}$, nonnegative volatility, positive ΔS , and one-dimensional spot/value vectors where required. The operator helper additionally validates that the spot grid is strictly increasing and uniformly spaced consistently with the supplied ΔS .

10 Testing Strategy

The test suite is designed to prove the local behavior of Ticket 2, not the future solver. The main categories are:

Test category	What it proves	What it does not prove
Spot grid tests	Endpoint, spacing, and length are correct.	Whether $S_{\max}$ is large enough for all option cases.
Tau grid tests	Endpoint, spacing, length, and $T = 0$ behavior are correct.	Any time-stepping accuracy.
Invalid grid tests	Bad grid sizes are rejected clearly.	A complete production input-validation policy.
Operator grid validation tests	Non-increasing grids, nonuniform grids, and dS -mismatched grids are rejected.	Any future nonuniform-grid method.
Interior-index tests	Boundary nodes are excluded from the PDE interior.	Correct future matrix assembly by itself.

Test category	What it proves	What it does not prove
Coefficient shape tests	Coefficient arrays align with interior nodes.	Correct time-step matrices.
Manual coefficient test	The lower, diagonal, and upper formulas match a hand calculation.	Full PDE-solver convergence.
Manufactured-function tests	$L_h(1) = -r$ and $L_h(S) = -qS$ hold on the grid.	Option-pricing accuracy for kinked payoffs.
Boundary helper tests	Simple put/call boundary formulas are implemented consistently.	Boundary right-hand-side adjustment in CN.

11 Test Results and Interpretation

The exact test command was:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src python3 -m unittest
discover -s tests
```

Final observed result:

```
Ran 25 tests in 0.002s
OK
```

This means the existing Ticket 1 tests and the new Ticket 2 grid/operator tests passed in the current Python runtime. It gives evidence that the local grid helpers, coefficient formulas, operator application, stronger operator-grid validation, and boundary-value helpers behave as expected for the tested cases.

It does not mean the CN solver is correct, because there is no CN solver yet. It also does not mean the American obstacle logic is correct, because there is no PSOR or American projection yet. The correct interpretation is narrow: Ticket 2 has a tested numerical setup layer for later solver validation.

12 Limitations

Ticket 2 has deliberate limits:

- no Crank-Nicolson time stepping;
- no PSOR;
- no American option solve;
- no linear complementarity solve;
- no boundary extraction;
- no Greeks;
- no experiments or generated result tables;
- no datasets, stress maps, or neural-network files;
- no proof that a chosen $S_{\max}$, M , or N is sufficient for production analysis.

The boundary helpers are only formulas for boundary values. They are not the complete boundary adjustments that a CN right-hand side will need. That distinction should remain visible in Ticket 3.

13 Lessons Learned / Study Notes

The first lesson is that a finite-difference method is mostly bookkeeping before it is linear algebra. If the grid and interior indices are wrong, a later solver can look sophisticated while still solving the wrong discrete problem.

The second lesson is that the Black-Scholes operator has financial meaning. Diffusion comes from volatility, drift comes from $r - q$, and discounting comes from $-rU$. The signs matter. A wrong sign in the drift term can reverse dividend behavior and damage later American call validation.

The third lesson is that boundary nodes are not ordinary solved nodes. They are imposed values that help the interior equations near the edge of the truncated domain. Later solver reports should continue to distinguish boundary behavior from interior PDE behavior.

The fourth lesson is that simple manufactured functions are useful. Checking $L_h(1) = -r$ and $L_h(S) = -qS$ is easier to understand than a full option price, but it still catches important operator mistakes.

The fifth lesson is to keep validation layered. Ticket 2 passing is good news for the grid and operator layer. It is not evidence that CN/PSOR, free-boundary extraction, or Greeks are correct.

14 Link to Ticket 3

Ticket 3 should build the European Crank-Nicolson validation mode. It can reuse:

- `uniform_spot_grid` and `uniform_tau_grid` for the numerical domain;
- `interior_indices` for the solved PDE nodes;
- `black_scholes_operator_coefficients` for the spatial operator;
- boundary helpers for European call and put validation cases;
- Ticket 1 closed-form prices as benchmark values.

Ticket 3 should compare finite-difference European prices against closed-form Black-Scholes prices before any American projection is added. It still should not implement PSOR, American option solving, boundary extraction, Greeks, experiments beyond the reviewed validation scope, datasets, stress maps, or neural networks unless a later approved ticket explicitly asks for them.

15 Reproducibility Record

15.1 Files Changed

Path	Status in this report task
<code>src/american_risk_surfaces/solvers/grid.py</code>	Created for grid and interior-index helpers; module docstring identifies Ticket 02.

Path	Status in this report task
src/american_risk_surfaces/solvers/operator.py	Created for operator coefficients and boundary helpers; module docstring identifies Ticket 02.
tests/test_grid_operator.py	Created for Ticket 2 unittest coverage.
reports/03_solver/tickets/ticket_02_grid_operator.tex	Created as the formal LaTeX report source.
reports/03_solver/tickets/ticket_02_grid_operator.pdf	Compiled from the LaTeX report source.

15.2 Commands

Test command:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src python3 -m unittest
discover -s tests
```

Compile command for Python syntax:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src python3 -m compileall
-q src tests
```

LaTeX compile command:

```
HOME=/private/tmp FC_CACHEDIR=/private/tmp/fontconfig-cache latexmk -xelatex -
interaction=nonstopmode -halt-on-error -outdir=/private/tmp/ticket02_latex
reports/03_solver/tickets/ticket_02_grid_operator.tex
```

PDF verification used `pdftinfo` and rendered page images with `pdftoppm`. The PDF is a LaTeX-generated formal report on A4 paper with one-inch margins, page numbers, tables, monospace commands and paths, and LaTeX math.